TESS planet candidate around TIC 147950620

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Summary

Signal found by the bls search of a public-data TESS pipeline (SPOC 120 s PDCSAP light curves, sectors 14, 20, 21, 40, 41, 47, 48, 74, 75). Final tier: **candidate: likely planet**. Catalogue cross-match (TOIs, CTOIs, published planets, SPOC TCEs, TESS eclipsing binaries): **TOI 1194.01** (TOI). CNN score (calibrated): 0.956. TRICERATOPS (10 Monte Carlo runs, mean ± error of the mean): FPP $= 0.0836 \pm 0.024$, NFPP $= 6.2e-24$.

Host star (TIC v8)

RA, Dec (deg)	167.82048, 69.964708
TESS magnitude	10.5
Radius ($R_\odot$)	0.993 ± 0.0558
$T_{\rm eff}$ (K)	5323 ± 124
Density ($\rho_\odot$)	0.938 ± 0.229
Distance (pc)	150

Transit fit

Limb-darkened model (batman, quadratic law, Kipping q_1, q_2), emcee with 48 walkers, 4000 steps (382 burn-in), run until the chain exceeded 50 autocorrelation times. Medians and 68% intervals.

Mid-transit time T_0 ($\mathrm{BJD_{TDB}}$)	$2459417.397891^{+0.000068}_{-0.000068}$
Period P (days)	$2.31064474^{+0.00000029}_{-0.00000030}$
Radius ratio $R_p/R_\star$	$0.0840^{+0.0018}_{-0.0015}$
Impact parameter b	$0.8321^{+0.0079}_{-0.0093}$
Stellar density $\rho_\star$ (g cm^{-3})	$1.567^{+0.086}_{-0.077}$
Limb darkening q_1	$0.333^{+0.090}_{-0.059}$
Limb darkening q_2	$0.46^{+0.33}_{-0.30}$
Depth (ppm)	6524^{+38}_{-37}
Duration T_{14} (hours)	$1.6224^{+0.0097}_{-0.0094}$
$a/R_\star$	$7.62^{+0.14}_{-0.13}$
Inclination (deg)	$83.73^{+0.18}_{-0.16}$
Planet radius ($R_\oplus$)	$9.11^{+0.54}_{-0.53}$
Semi-major axis (au)	$0.0352^{+0.0021}_{-0.0021}$
Equilibrium temperature (K, $A = 0$)	1364^{+34}_{-34}
Insolation ($S_\oplus$)	576^{+60}_{-56}

Vetting

Test	Value	Pass condition	Result
Transit SNR (correlated noise)	217	≥ 7.1	pass
Transits observed	87	≥ 2 (periodic)	pass
Odd/even depth difference (σ)	1.85	≤ 4	pass
Secondary eclipse (σ)	1.85	≤ 3.7	pass
Ingress fraction (V shape)	0.244	≤ 0.4	pass
Planet radius ($R_\oplus$)	8.62	≤ 25	pass
Centroid source offset (arcsec)	2.88	4σ and $>21''$	pass
Difference image offset (arcsec)	1	4σ and $>10.5''$	pass
Difference image significance (σ)	0.346	–	–
CCD common mode (σ)	0.174	≤ 4	pass
Crowding (CROWDSAP)	0.98	≥ 0.8	pass
Stellar density ratio (max/TIC)	4.63	≥ 0.2	pass

Caveats

- This is a candidate with statistical evidence, not a confirmed planet: confirmation needs high-resolution imaging, reconnaissance spectroscopy and radial velocities.
- TRICERATOPS was run without a high-resolution imaging contrast curve, so close bound companions are not excluded.
- The MCMC converged (chain longer than 50 autocorrelation times, stable estimate) (acceptance 0.24, 4000 steps, about 2267 independent samples); errors are scaled by the time-averaging factor $\beta = 1.90$ for correlated noise.
- Stellar radius and effective temperature come from TIC v8; if the TIC gives no error, 7% (radius) and 150 K were assumed.

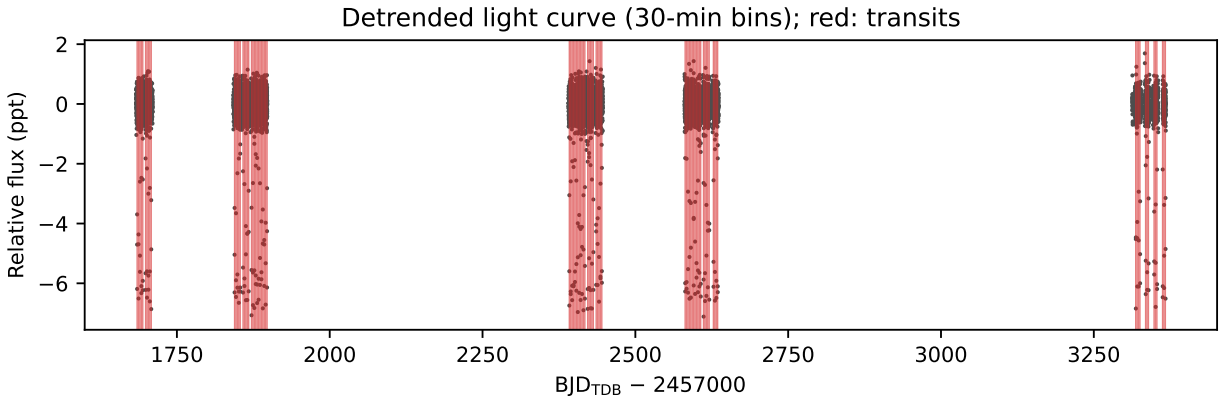


Figure 1: Detrended SPOC PDCSAP light curve (transit-protected biweight).

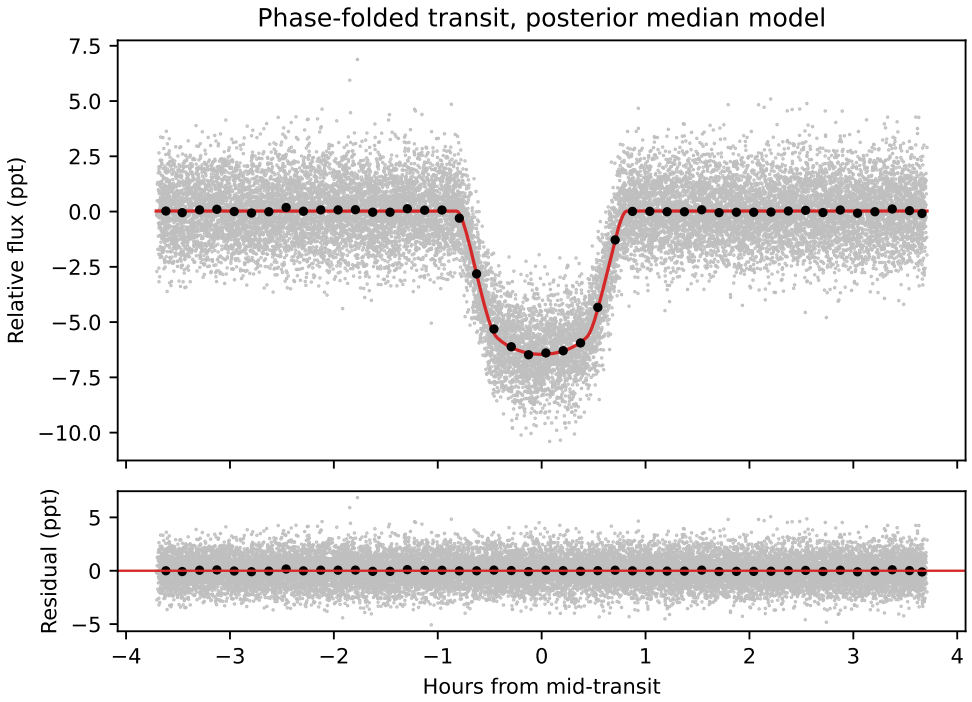


Figure 2: Phase-folded transit with the posterior median limb-darkened model.

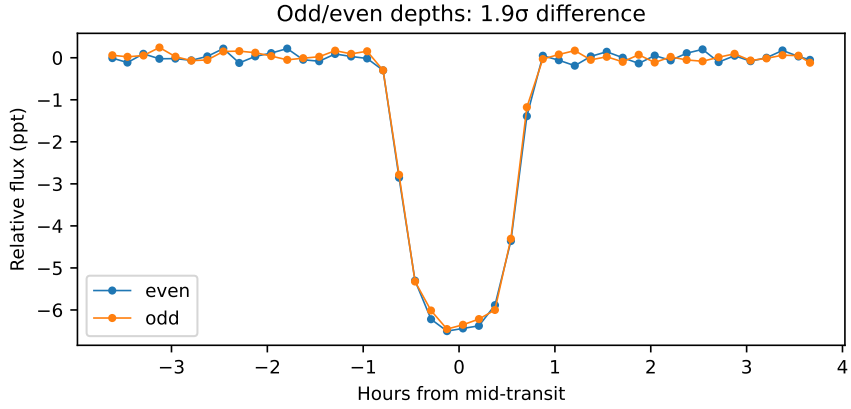


Figure 3: Odd and even transits, binned to 10 minutes.

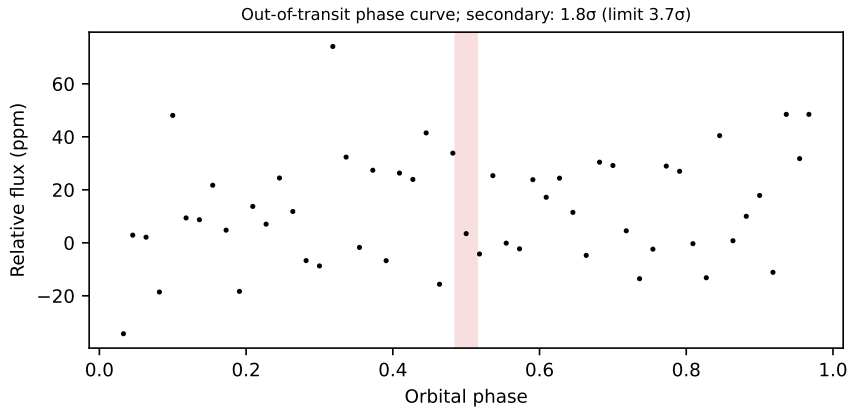


Figure 4: Phase curve; the shaded band is the secondary eclipse window for a circular orbit.

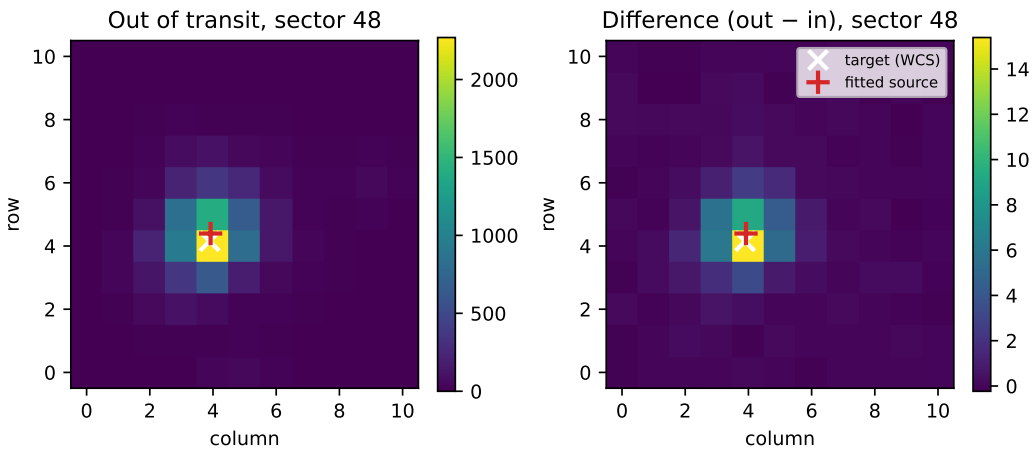


Figure 5: SPOC target pixel file: out-of-transit image and difference image. White cross: target catalogue position; red plus: fitted source.

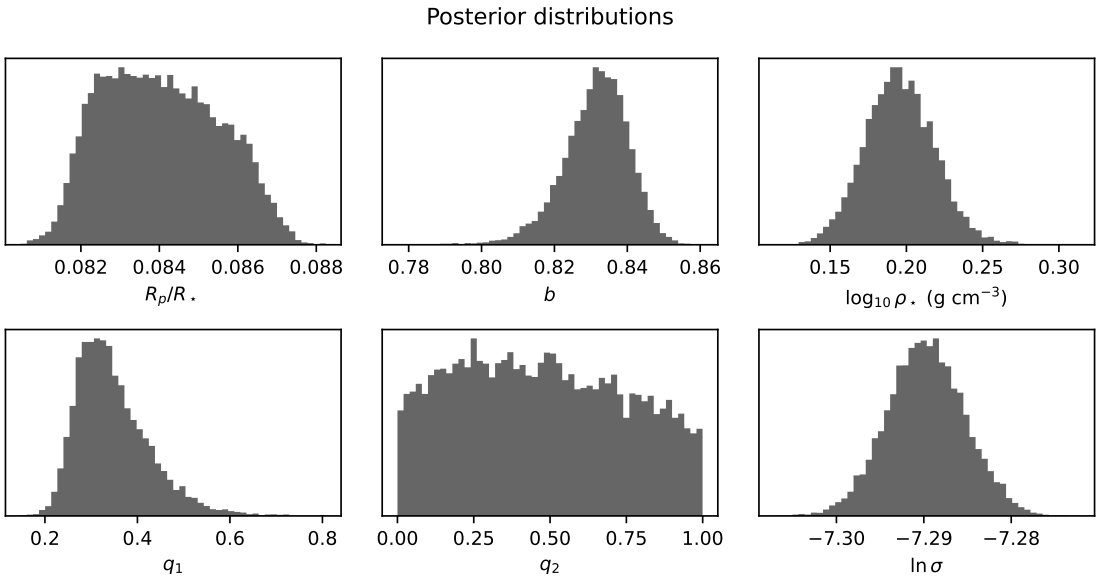


Figure 6: Posterior distributions of the fitted parameters.